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Comparative analysis of astrocytes in the prefrontal cortex of primates: Insights into the evolution of human brain energetics

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Abstract

Astrocytes are the main homeostatic cell of the brain involved in many processes related to cognition, immune response, and energy expenditure. It has been suggested that the distribution of astrocytes is associated with brain size, and that they are specialized in humans. To evaluate these, we quantified astrocyte density, soma volume, and total glia density in layer I and white matter in Brodmann's area 9 of humans, chimpanzees, baboons, and macaques. We found that layer I astrocyte density, soma volume, and ratio of astrocytes to total glia cells were highest in humans and increased with brain size. Overall glia density in layer I and white matter were relatively invariant across brain sizes, potentially due to their important metabolic functions on a per volume basis. We also quantified two transporters involved in metabolism through the astrocyte-neuron lactate shuttle, excitatory amino acid transporter 2 (EAAT2) and glucose transporter 1 (GLUT1). We expected these transporters would be increased in human brains due to their high rate of metabolic consumption and associated gene activity. While humans have higher EAAT2 cell density, GLUT1 vessel volume, and GLUT1 area fraction compared to baboons and chimpanzees, they did not differ from macaques. Therefore, EAAT2 and GLUT1 are not related to increased energetic demands of the human brain. Taken together, these data provide evidence that astrocytes play a unique role in both brain expansion and evolution among primates, with an emphasis on layer I astrocytes having a potentially significant role in human-specific metabolic processing and cognition.

KEYWORDS
astrocyte, EAAT2, GFAP, glia, GLUT1, human, prefrontal cortex, primate

1 | INTRODUCTION

The human cerebral cortex is characterized by the upregulation of metabolic genes and increased energy utilization compared to other primates (Bauernfeind et al., 2015; Cáceres et al., 2003; Grossman et al., 2004; Khaitovich et al., 2006; Preuss et al., 2004; Sokoloff, 1960; Uddin et al., 2004). Additionally, lactate dehydrogenase isoenzymes, which play an important role in processing glucose from food, are altered in the primate lineage, likely evolving to meet the increased energy requirements of the enlarged brains of primates (Duka et al., 2014). Astrocytes play a role in regulating central nervous system (CNS) metabolism via the astrocyte–neuron lactate shuttle, which couples synaptic activity and energy delivery (Magistretti et al., 1999; Magistretti, 2006; Magistretti & Pellerin, 1999; Pellerin et al., 1998; Pellerin & Magistretti, 1994, 1997).

The astrocyte–neuron lactate shuttle allows astrocytes to link neuronal excitation to glucose uptake (Magistretti et al., 1999; Magistretti, 2006; Magistretti & Pellerin, 1999; Pellerin et al., 1998; Pellerin

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& Magistretti, 1994, 1997). In brief, glutamate is released at the synapse, activating glutamatergic receptors on the postsynaptic cell. Most synaptic glutamate is taken up by astrocytes via the excitatory amino acid transporter 2 (EAAT2), which disrupts sodium homeostasis (Pellerin & Magistretti, 1997). Re-establishment of sodium homeostasis and formation of glutamine from glutamate decreases adenosine triphosphate (ATP) stores within the astrocyte, creating a metabolic burden that triggers glucose uptake from the circulation through glucose transporter 1 (GLUT1). GLUT1 is expressed by capillary endothelial cells and astrocytes (Magistretti & Allaman, 2015; Pellerin et al., 2007; Simpson et al., 2007; Winkler et al., 2015). As glucose is broken down, pyruvate is formed and converted to lactate by lactate dehydrogenase 5, which is primarily expressed by astrocytes; lactate is then shuttled to neurons (Halestrap, 2013; Pierre & Pellerin, 2005). Therefore, astrocytes are integral for the maintenance of neuronal energy requirements and have the ability to dramatically upregulate glycolysis, whereas neurons cannot (Y. Zhang et al., 2014).

Dysfunction in GLUT1 in astrocytes leads to severe neurological phenotypes, while enhancing GLUT1 is neuroprotective (Bliss et al., 2004; Wang et al., 2006). Similarly, dysregulation of EAAT2 has been linked to excitotoxicity, neuronal death, and neurological disorders such as Alzheimer's disease, Parkinson's disease, and schizophrenia (Karki et al., 2013, 2015; Kornhuber & Wiltfang, 1998; Takahashi et al., 2015; Young et al., 2014). Therefore, examining two of the important transporters, EAAT2 and GLUT1, involved in the astrocyte–neuron lactate shuttle could shed light on unique properties of human astrocytic function related to metabolism and brain health (Bauernfeind et al., 2015; Cáceres et al., 2003; Grossman et al., 2004; Khaitovich et al., 2006; Preuss et al., 2004; Uddin et al., 2004).

Astrocytes are also important in the brain's inflammatory response (Anderson et al., 2014; Heneka et al., 2010; Markiewicz & Lukomska, 2006; Sofroniew, 2009; Sofroniew & Vinters, 2010). Astrogliosis is the process by which astrocytes react to trauma, infection, and neurodegeneration, and it includes a spectrum of molecular, cellular, and functional changes marked by the upregulation of an astrocyte-specific intermediate filament called glial fibrillary acidic protein (GFAP), hypertrophy of the soma and cellular processes, and loss of domain organization (Pekny et al., 2014; Sofroniew, 2009; Sofroniew & Vinters, 2010; Wilhelmsson et al., 2006). Species differences have been noted in astrogliosis. Cortical astrocytes in chimpanzees do not demonstrate a widespread reactive phenotype in response to aging and neuropathology, which differs from humans (Munger et al., 2019). Furthermore, accumulating evidence suggests that a difference exists in the inflammatory response to brain pathology and aging between humans and nonhuman primates (Autrey et al., 2014; Edler et al., 2018; Munger et al., 2019; Sherwood et al., 2011).

Glia to neuron (G:N) ratios increase with brain size in certain regions of the neocortex across mammalian species (Friede, 1954; Haug, 1987; Hawkins & Olszewski, 1957; Sherwood et al., 2006; Stolzenburg et al., 1989; Tower, 1954), with the interpretation that this increase most likely reflects a greater neuronal energy expenditure, increased activity of synaptic transmission, and a higher demand for homeostatic clearance (Attwell & Laughlin, 2001; Sherwood et al., 2006). The human frontal cortex possesses an overall increased G:N ratio relative to other primates, coinciding with allometric expectations based on overall brain size (Sherwood et al., 2006). Furthermore, the frontal cortex of primates is disproportionately enlarged relative to other mammals, and specifically in humans, contains more neuropil space for synapses and dendritic arbors (Bush & Allman, 2004; Garin et al., 2022; Semendeferi et al., 2001, 2002, 2011; Smaers et al., 2010; Spocter et al., 2012). This region also displays more extreme vulnerability to brain aging in humans relative to other primates, as seen in volumetric and behavioral studies (Autrey et al., 2014; Pfefferbaum et al., 1998; Raz et al., 2005, 2007; Resnick et al., 2003; Scahill et al., 2003; Sherwood et al., 2011). Specifically, while an overall volumetric decrease was not observed in the prefrontal cortex of macaque monkeys, subregions displaying age-related effects impacting cognition and microstructural changes have been noted in aging animals that include a loss of dendritic spines in cortical pyramidal neurons (Dumitriu et al., 2010; Luebke et al., 2015). A recent study on chimpanzees also demonstrated a unique vulnerability of interlaminar astrocytes to Alzheimer's disease (AD)-type pathology in the prefrontal cortex (Brodmann's areas 9 and 10; Munger et al., 2019). This included increased astrogliosis in layer I that was positively associated with increasing amyloid beta and tau protein expression. An increase in reactive astrocytes in the presence of AD pathology is commonly observed in human AD brains. This follows previous work that showed a unique disruption of astrocyte morphology to pathology (Colombo et al., 2002). Interlaminar and fibrous astrocytes are the two subtypes of astrocytes found in all mammals and are immunoreactive for GFAP (Ramón y Cajal, 1897; Falcone et al., 2019; Oberheim et al., 2006, 2009). Interlaminar astrocyte cell somas are found in layer I of the cerebral cortex, and they extend their processes into layers II through IV (Colombo et al., 1995, 2004; Colombo & Reisin, 2004; Falcone et al., 2019; Oberheim et al., 2009). Fibrous astrocytes are found in the white matter (WM) and have processes that surround blood vessels (Ramón y Cajal, 1897; Oberheim et al., 2006, 2009). Both astrocyte subtypes have been reported to be more numerous, larger, and more complex in structure in primates than in other mammals (Falcone et al., 2019; Oberheim et al., 2009).

Our goal in this study was to quantify the overall numbers of astrocytes and the expression of EAAT2 and GLUT1 within the dorsolateral prefrontal cortex (Brodmann's area 9) across several primate species to provide a comparative context to understand human brain energetics.

2 | MATERIALS AND METHODS

2.1 | Specimens

The study sample included humans (*Homo sapiens*), chimpanzees (*Pan troglodytes*), olive baboons (*Papio anubis*), and pigtailed macaques (*Macaca nemestrina*), with a minimum of six individuals per species with sexes balanced (Table 1). All nonhuman brain samples were obtained from Association for Assessment and Accreditation of Laboratory Animal Care (AAALAC) or American Zoos and Aquariums (AZA) accredited zoological or biomedical research facilities and were

TABLE 1 Individuals included in the present study with age (years), sex, and brain weight (grams) listed. When the exact brain weight for a specimen was unknown, it was extrapolated from one hemisphere (*) or the average brain weight for the species was used (**, Robson & Wood, 2008)

Genus, species	Age (years)	Sex	Brain weight (g)
Macaca nemestrina	3	M	102.1
	4	M	118.0
	6	F	82.4
	9	F	79.1
	15	F	103.3
	16	M	102.1
Papio anubis	5	F	158.7
	6	M	124.2*
	7	M	182.4
	9	M	181.6
	10	F	181.2
	10	M	158.5
Pan troglodytes	24	M	375.5
	35	F	302.2
	40	F	371.5
	40	M	341.2
	41	M	384**
	43	F	356.8
	51	F	387.7
Homo sapiens	37	F	1181
	37	M	1620
	38	F	1210
	39	M	1650
	42	F	1380
	44	M	1608

housed in accordance with each institution's animal care and use guidelines. Macaque brain specimens were acquired from the Washington National Primate Research Center, and baboon brain samples were obtained from the Southwest National Primate Research Center. All chimpanzee brain specimens were provided by the National Chimpanzee Brain Resource. Nonhuman individuals died of natural causes or were humanely euthanized for quality of life. Human brain samples were acquired through the NIH NeuroBioBank's Brain and Tissue Repository. All individuals were adult and unaffected controls (i.e., absence of neuropathology). It should be noted that we initially examined six young chimpanzees for this study that had to be removed from the sample due to the presence of global inflammation as indicated by severe astrogliosis. This resulted in a chimpanzee sample that included a wider age range relative to the other species. To ensure that age was not a confounding factor for these analyses, we evaluated the relationship between age and each variable (for details see below) within chimpanzees. Only one of the 11 variables under investigation correlated with age (glia density in layer I, Spearman's rho = −0.78, $p = .048$), indicating that the age range within the chimpanzee sample is not a confounding factor for this study.

2.2 | Sample processing

The regions of interest were layer I of prefrontal cortex Brodmann's area 9 and the WM adjacent to it. Area 9 is located on the dorsal region of the lateral prefrontal cortex and extends medially to the paracingulate sulcus of humans and the cingulate sulcus of macaque monkeys (Paxinos et al., 2000; Petrides & Pandya, 1999). The cytoarchitecture of area 9 is recognizable by a layer IV that is narrow and not well-developed, a compact layer II and upper layer III. The lower portion of layer III contains large pyramidal neurons and is not cell dense (Petrides & Pandya, 1999). Area 9 was easily distinguished from area 10 (rostral to area 9), because area 10 has a well-developed layer IV and an overall pale appearance due to a sparser distribution of cells (Semendeferi et al., 2001). Layer I and the WM are easily identified on GFAP-immunostained sections, as these regions have the highest density of immunoreactive astrocytes. Brains were collected (postmortem interval <20 h) and immersion fixed in 10% buffered formalin for a minimum of 7 days. Brains were then transferred to a 0.1 M phosphate buffered saline solution with 0.1% sodium azide and stored at 4°C. Prior to sectioning, all brain samples were cryoprotected in a series of graded sucrose solutions (10%, 20%, and 30%) until saturated. Samples were quickly frozen with dry ice and sectioned using a freezing sliding microtome at a thickness of 40 μm (SM2000R; Leica, Chicago, IL, USA). Each section was placed in a microcentrifuge tube containing freezer storage solution (30% each distilled water, ethylene glycol, and glycerol and 10% 0.244 M PBS), numbered sequentially, and stored at −20°C until further histological or immunohistochemical processing. A 1-in-10 series of sections was taken from each sample and stained for Nissl substance using a 0.5% cresyl violet to allow for the visualization of cytoarchitecture.

2.3 | Antibody characterization

Table 2 provides details about the antibodies used in this study. GFAP was detected using a rabbit polyclonal antibody raised against glial fibrillary acidic protein (1:12,500 dilution, Millipore, Burlington, MA, USA, Cat# AB5804, RRID: AB_2109645). This antibody stains cells that are consistent in morphology and distribution with previous reports (e.g., Colombo, 1996; Falcone et al., 2019; Hansen et al., 1987). EAAT2 was visualized using a guinea pig polyclonal antibody from the carboxy terminus of rat EAAT2 (1:3,000 dilution, Millipore, CAT# AB1783, RRID: AB_90949). Preabsorption of the antiserum with the immunogen peptide (Cat# AG91) completely abolishes the staining (manufacturer's datasheet). Staining is consistent with previous reports (e.g., Roberts et al., 2014). GLUT1 was detected using a mouse monoclonal antibody that corresponds to the carboxy terminus of human GLUT1 (1:5,000 dilution, Abcam, CAT# ab40084, RRID: AB_2190927). GLUT1 antibody

TABLE 2 Primary antibodies used for immunohistochemistry

Antigen	Antibody	Host/type	Dilution	Supplier/catalog number	RRID
GFAP	Glial fibrillary acidic protein	Rabbit, polyclonal	1:12,500	Millipore, AB5804	AB_2109645
EAAT2	Synthetic peptide from the carboxy-terminus	Guinea pig, polyclonal	1:3,000	Millipore, AB1783	AB_90949
GLUT1	Glucose transporter 1	Mouse, monoclonal	1:5,000	Abcam, ab40084	AB_2190927

staining was consistent with what has been previously reported (e.g., Leiter et al., 2019).

2.4 | Immunohistochemistry

A series of brain sections from each region were immunohistochemically processed using the avidin-biotin-peroxidase method to detect the expression of GFAP, EAAT2, and GLUT1 (Beach et al., 1989; Hansen et al., 1987; Munger et al., 2019; Rodríguez et al., 2014; Sofroniew & Vinters, 2010). Sections were also double-labeled for GFAP and GLUT1 to visualize astrocytes interacting with GLUT1-ir vasculature. Tissue sections were first pretreated for antigen retrieval for 30 min in a 0.05% solution of citraconic acid (pH 7.4) at 85°C except sections immunostained for EAAT2, which did not undergo an antigen retrieval step. Endogenous peroxidases were quenched for 20 min in a solution of 75% methanol, 2.5% hydrogen peroxide (30%), and 22.5% distilled water. Sections stained for GFAP were preblocked in a solution containing 90.4% 0.1 M PBS (pH 7.4), 0.6% Triton X-100, 4% normal serum, and 5% bovine serum albumin. The preblock solution for EAAT2 sections contained 86% 0.1 M PBS (pH 7.4), 4% normal serum, 5% bovine serum albumin, and 5% dried milk. Tissue was incubated in a primary antibody for 24 h at room temperature (RT) followed immediately by 24 h at 4°C (48 h for EAAT2; Table 2). After incubation, sections were placed in a solution of 2% normal serum and biotinylated secondary antibody at varying concentrations (1:2,500 for EAAT2 and 1:200 for GFAP). Sections were placed in avidin-peroxidase complex (PK-6100; Vector Laboratories, Burlingame, CA, USA) followed by a 3,3′-diaminobenzidine-peroxidase substrate with nickel enhancement for visualization (SK-4100; Vector Laboratories). Sections immunostained for GFAP were incubated in a blocking solution containing 5% normal serum and 95% 0.1 M PBS. The tissue was incubated for 48 h in anti-GLUT1 antibody (24 h at RT followed by 24 h at 4°C). Following incubation, sections were placed in a solution of biotinylated secondary antibody (1:200) and 2% normal serum. Sections were incubated in avidin-peroxidase complex followed by NovaRED (SK-4800; Vector Laboratories).

2.5 | Data acquisition

Stereology was performed using an Olympus BX51 photomicroscope equipped with a digital camera and StereoInvestigator software (MBF Bioscience, Williston, VT, USA). Nissl-stained sections adjacent to the immunostained sections were used to guide delineation of cortical layers at low magnification (4X, 0.13 N.A.; Figure 1). Stereology parameters were initially determined for each species to ensure coefficients of error below 0.10, and the mean number of sampling sites and markers counted per individual are reported in Table 3. The optical fractionator probe was used to quantify (1) total glia density (Gv) in layer I and WM of Nissl-stained sections at high magnification (60X, 1.42 N.A.) with counting frames of 75 × 75 μm, an optical disector height of 10 μm, and a 2% guard zone; (2) GFAP-ir astrocyte density (GFAPv) in layer I and WM at 40X, 0.75 N.A., with counting frames of 100 × 100 μm, an optical disector height of 10 μm, and a 2% guard zone; and (3) EAAT2-ir astrocyte density (EAAT2v) in WM at 60X, 1.42 N.A., with counting frames of 50 × 50 μm, an optical disector height of 8 μm, and a 2% guard zone. Soma volume (SV) was measured for GFAP-ir and EAAT2-ir astrocytes using the nucleator probe (six rays) during the optical fractionator probe, measuring every second EAAT2-ir or

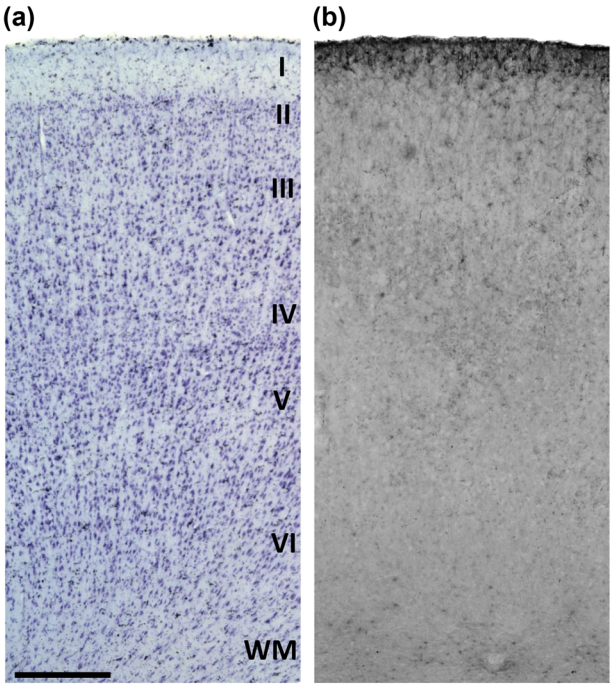


FIGURE 1 Nissl-stained (a) and glial fibrillary acidic protein (GFAP)-immunostained (b) sections with cortical layers labeled in prefrontal cortex area 9 of a chimpanzee. Nissl-stained sections were used to denote the appropriate cortical layers (layer I and white matter) prior to quantification for each sample. Scale bars = 250 μm.

TABLE 3 Average sampling sites, number of markers counted, and coefficient of error (Gunderson $m = 1$ for densities) for the variables included in this study

Variable	Sampling sites per individual (mean ± standard deviation)	Number of markers counted per individual (mean ± standard deviation)	Coefficient of error
Gv Layer I	70 ± 9	806 ± 255	0.05
Gv WM	75 ± 6	1,605 ± 382	0.04
GFAPv Layer I	127 ± 16	170 ± 66	0.09
GFAPv WM	129 ± 12	212 ± 114	0.08
GFAP SV Layer I	127 ± 16	61 ± 25	<0.01
GFAP SV WM	129 ± 12	70 ± 38	<0.01
EAAT2v	74 ± 7	255 ± 52	0.07
EAAT2 SV	74 ± 7	121 ± 26	<0.01
GLUT1 Vol	63 ± 7.3	1,287 ± 396	0.03
GLUT1 AF (%)	63 ± 7.3	1,287 ± 396	0.03

Abbreviations: AF, area fraction; GFAP, glial fibrillary acidic protein; GLUT1, glucose transporter 1; GV, glia density; EAAT2, excitatory amino acid transporter 2; SV, soma volume; v, density; Vol, volume; WM, white matter.

third GFAP-ir astrocyte. For all optical fractionator probes, the section thickness was measured every fifth sampling site. Densities (mm^3) were calculated as the total population divided by planimetric measurement of the reference volume. For comparison among species, Gv was used as the denominator.

The volume of vasculature immunoreactive for GLUT1 was measured in the gray matter using the area fraction fractionator probe (AFF), a Cavalieri point counting system. Using a 10X objective (0.25 N.A.), markers were placed on a grid of points (300 μm × 300 μm) overlaying the sampling area. Every point on the grid received one of two markers: non-GLUT1 vessel or GLUT1 vessel. The estimated area fraction (AF) of GLUT1-ir vessels was calculated by the AFF probe and presented as a percentage of the total area sampled. AF refers to the fraction of the tissue occupied by the subregion (GLUT1-ir vessels). Additionally, an estimated volume of GLUT1-ir vessels was reported and divided by the planimetric volume of the region to provide an estimate of the total volume of GLUT1-ir vessels (mm^3) in the region.

2.6 | Statistical analyses

Independent samples *t*-tests were used to check for sex differences within species for each measurement. A Levene's test for equality of variances was performed to determine that the variances were equal between both sexes for each *t*-test completed. This assumption was rejected for multiple *t*-tests, and for those analyses, the reported statistic assumes unequal variances. Statistical analyses were performed using SPSS (version 24) and Statistica (version 13.3). Alpha was set at 0.05, and a Bonferroni adjustment was used for multiple comparison testing. To examine the contribution of astrocytes to the increased G:N ratio with brain size, linear regression analyses were used to evaluate Gv, GFAPv, GFAPv/Gv, and GFAP-ir astrocyte soma volume (GFAPSV) in both layer I and the WM against brain weight. Brain weight in grams was transformed to natural log (ln) prior to performing regression analyses. A mixed model analysis of variance (ANOVA) with repeated measures was used to examine Gv, GFAPv, GFAPv/Gv, and GFAPSV in layer I and the WM among the primate species: macaque, baboon, chimpanzee, and human. Region (layer I and WM) was the within-subjects factor, and species was the between-subjects factor. Fisher's least significance difference (LSD) post hoc tests were performed to evaluate significant effects. One-way ANOVA were used to examine each of the remaining variables (EAAT2v, EAAT2/Gv, EAAT2sv, GLUT1-ir vessel volume, and GLUT1 AF) among species. If the test of homogeneity of variances using Levene's statistic was significant, a nonparametric Kruskal–Wallis test was performed. Fisher's LSD post hoc tests were used to evaluate significant effects.

3 | RESULTS

Both layer I and the WM of area 9 are rich in glia cells, and the presence of neurons is limited (Figures 2 and 3). Overall, glia were more densely packed in the WM relative to layer I in all species (Figure 3). GFAP-ir interlaminar and fibrous astrocytes were present in all species (Figures 4–7). Fibrous astrocytes had an irregularly shaped soma with numerous processes (Figure 4). As previously reported, these cells appeared disorganized and did not exist in neat domains (Figure 4). The astrocytic processes in all nonhuman primate species appeared smooth; however, processes extending from human fibrous astrocytes displayed bead-like varicosities. (Figure 4). Interlaminar astrocytes are shown in Figure 5, and all primates exhibited the classic interlaminar palisade (Figures 5 and 6), with numerous processes that extended in a columnar fashion toward the WM and terminating between layers II and IV (Figure 6).

Closer examination revealed variation among species (Figure 7). Macaques and baboons displayed twisting-like processes but with a

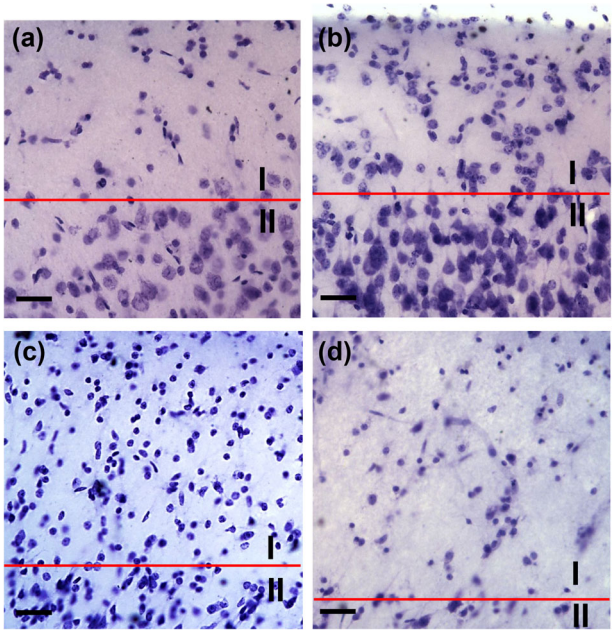


FIGURE 2 Nissl-stained photomicrographs of layer I of the prefrontal cortex area 9 of a pigtailed macaque (a), olive baboon (b), chimpanzee (c), and human (d). Macaques and baboons (panels a, b) have a thin layer I, and a red line is used to demarcate the transition to layer II. Cortical layers are marked in panels a-d, demonstrating the distinction between layers I and II. Layer I is distinguished by the dominance of glial cells. Scale bars = 25 μm.

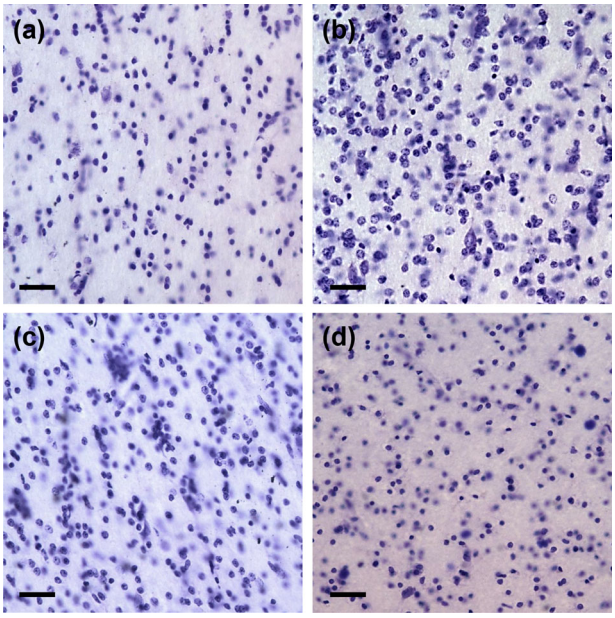


FIGURE 3 Nissl-stained photomicrographs of prefrontal cortex area 9 white matter in pigtailed macaque (a), olive baboon (b), chimpanzee (c), and human (d). The white matter is a region dominated by glial cells, and images are taken in deeper white matter (i.e., not immediately adjacent to layer VI). Astrocytes and oligodendrocytes were quantified together as "glial cells" as it is not always possible to distinguish each using Nissl-stained sections. Scale bars = 25 μm.

greater density, and their palisades passed through more cortical layers before terminating (Figure 7). However, it appeared that not all chimpanzees had interlaminar astrocyte palisades (Figure 7). A loss of interlaminar palisades has been reported due to pathology and astrocyte activation that is associated with inflammation (Figure 7k; Colombo et al., 2002; Colombo & Reisin, 2004; Munger et al., 2019). Notably, even chimpanzees without pathology and astrogliosis lacked the interlaminar palisade (Figure 7). Interlaminar astrocytes in humans and chimpanzees, when present, had varicosities (Figure 7).

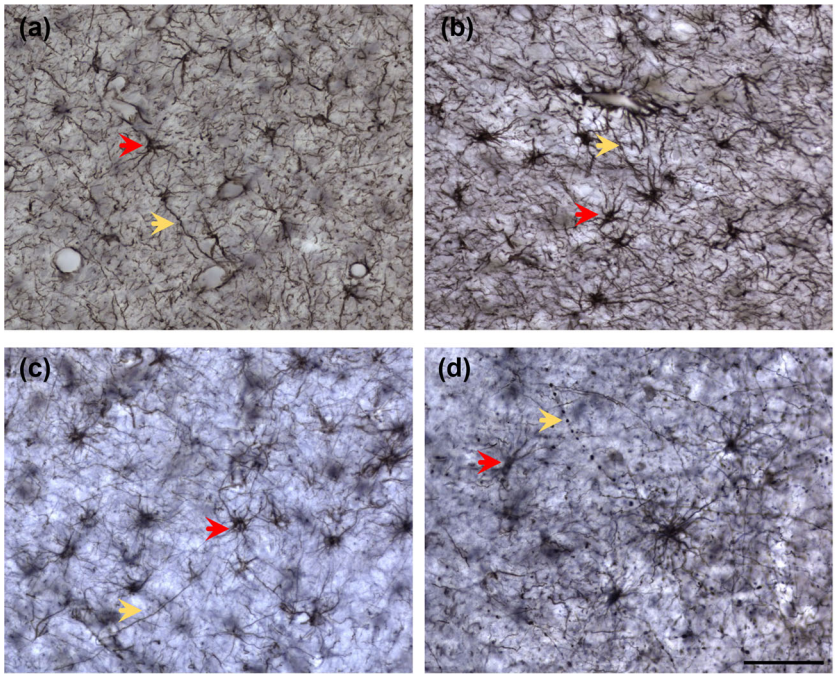


FIGURE 4 Glial fibrillary acidic protein (GFAP)-stained sections of prefrontal cortex area 9 white matter (WM) of pigtailed macaque (a), olive baboon (b), chimpanzee (c), and human (d). One example of a GFAP-ir astrocyte (red arrow) and process (yellow) are indicated in each panel. Note that fibrous astrocytes in every species have irregular soma shapes with several extending processes. These astrocytes also have overlapping domains with astrocytes existing in a "chaotic" or nonpatterned layout throughout the WM in each species. Lastly, processes extending from fibrous astrocytes have a smooth or uninterrupted appearance in every species except humans (c). Human processes appear dotted or varicosity-like and have a similar appearance to the processes of interlaminar astrocytes. Scale bar = 50 μm.

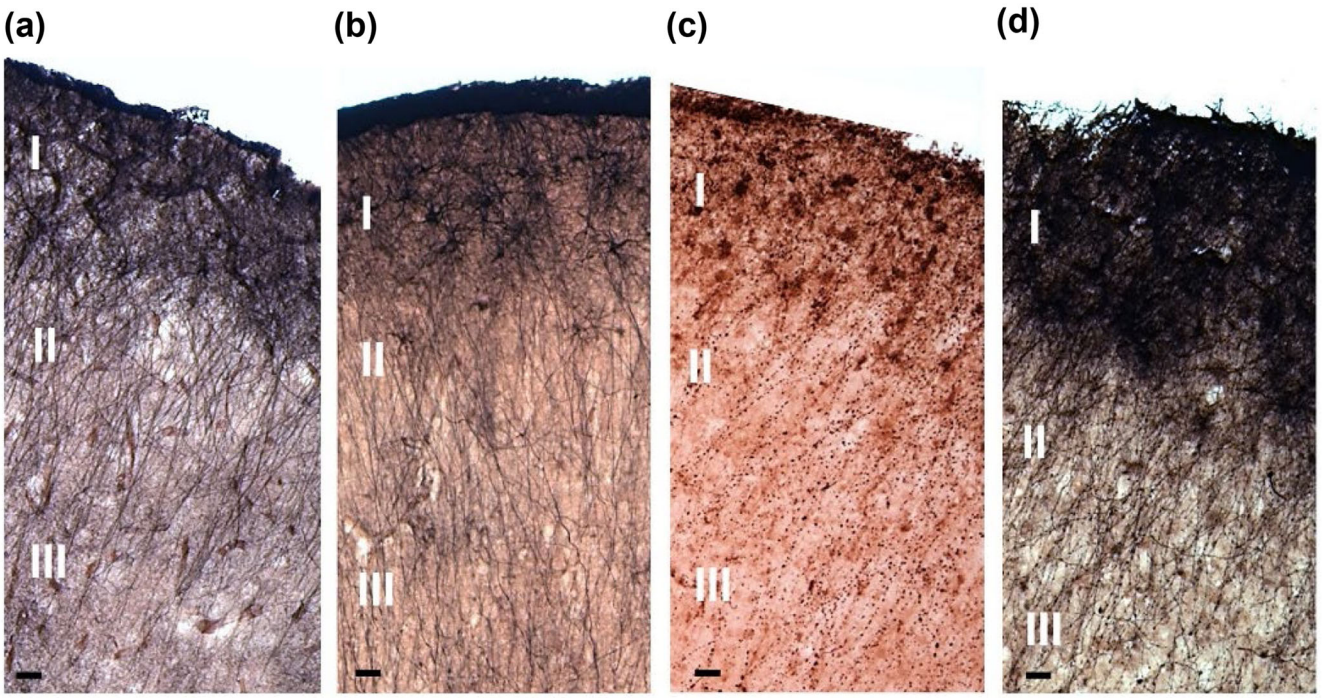


FIGURE 5 Immunohistochemically processed sections for glial fibrillary acidic protein (GFAP) showing the upper cortical layers of the frontal cortex of pigtailed macaque (a), olive baboon (b), chimpanzee (c), and human (d). Cortical layers (I–III) are labeled on each panel. Interlaminar astrocyte cell bodies are found in layer I and are often described as having several long torturous processes extending into deeper cortical layers (II–IV). Human and chimpanzees are the only two species to demonstrate interlaminar processes with a bead-like appearance that are putative varicosities (c and d). Scale bars = 25 μm.

These varicosities occurred at regular intervals along the processes as they extended toward deeper cortical layers. Lastly, it appeared that humans had the greatest number of processes and varicosities (Figure 7).

Additionally, astrocyte processes could be visualized extending upward toward the pia near the junction of layer VI and the WM (Figure 8). Previous studies have reported that the presence of polarized astrocytes is unique to primates (Oberheim et al., 2006). These astrocytes are found near the WM in layers V and VI and extend one or two long processes toward layer I (Oberheim et al., 2006). Here, we also report the presence of polarized astrocytes (Figure 9). We observed that the processes are numerous and appear to mostly come from cell bodies adjacent to, or in, the WM (Figure 8). The processes extending from these cell bodies twist, giving them a tortuous appearance. While most of these processes extend toward the pia, the processes often have an irregular pattern of directionality (Figure 8). Varicosities were regularly found on these upward projecting processes in humans (Figure 8) and occasionally in other species (Figure 8). The presence of polarized astrocytes containing varicosities likely represents varicose projection astrocytes, which have previously been reported in all hominoids but not in other primates (Falcone et al., et al., 2022; Oberheim et al., 2009). These polarized astrocytes possess varicosities at equally spaced intervals (Oberheim et al., 2009). Most of these processes terminated within layers V and VI, although some processes extended further upward.

3.1 | Relationship of glia and astrocytes with brain weight

Summary data for all variables are shown in Table 4. Regression analyses found that Gv in layer I and WM were not associated with brain weight in this sample of catarrhine primates (layer 1: $F_{1,24} = 1.86$, $p = .19$, $R^2 = 0.08$; WM: $F_{1,24} = 0.00$, $p = .95$, $R^2 = 0.00$; Figure 9a,b). GFAPv in layer I increased ($F_{1,24} = 4.77$, $p = .04$, $R^2 = 0.17$; Figure 9c) but was not significant when correcting for multiple comparisons. GFAPv in the WM did not vary significantly in relation to brain weight ($F_{1,24} = 4.66$, $p = .42$, $R^2 = 0.17$; Figure 9d).

The ratio of GFAP/Gv in layer I increased with brain weight ($F_{1,24} = 15.10$, $p = .001$, $R^2 = 0.40$; Figure 10a), whereas WM GFAP/Gv was not different across brain size variations ($F_{1,24} = 2.46$, $p = .13$, $R^2 = 0.10$; Figure 10b). GFAP-ir SV in layer I significantly increased with brain weight ($F_{1,24} = 23.37$, $p = .00$, $R^2 = 0.50$; Figure 10c), but there was no relationship in the WM ($F_{1,24} = 3.12$, $p = .09$, $R^2 = 0.12$; Figure 10d).

3.2 | Glia and GFAP-ir astrocyte profiles

There were no differences between sexes for any of the variables; therefore, sexes were pooled for all analyses (all p values > .006). The mixed model ANOVA examining Gv by region (layer I or WM) in the

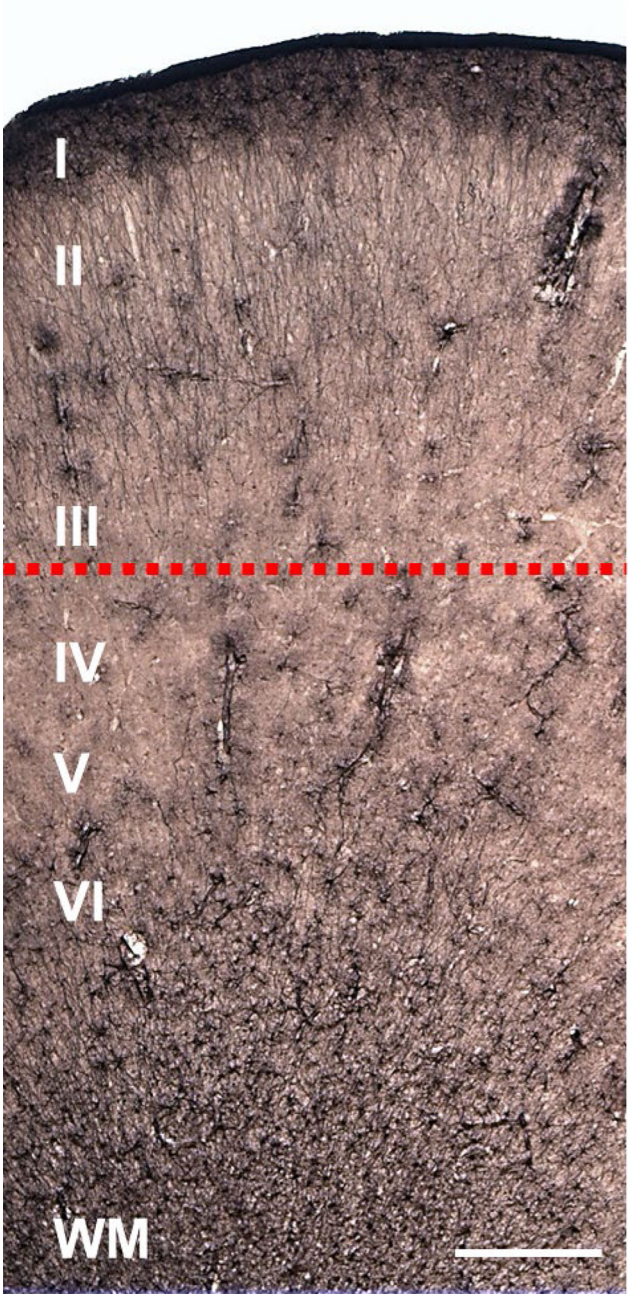


FIGURE 6 Baboon prefrontal cortex area 9 immunostained for glial fibrillary acidic protein (GFAP) with labeled cortical layers. Interlaminar astrocyte cell bodies are located in layer I, although at this magnification, they are hard to distinguish. These astrocytes project several long processes into deeper cortical layers. In catarrhines primates, they extend in a columnar or palisade fashion toward layers III and IV. The dashed red line indicates the point where the majority of interlaminar processes have terminated. Scale bar = 250 μm.

four primates showed significant main effects of region ($F_{1,21} = 25.58$, $p < .01$) and species ($F_{3,21} = 3.28$, $p = .04$) and a significant interaction between region and species ($F_{3,21} = 5.49$, $p < .01$; Figure 11a). Fisher's LSD post hoc testing revealed that Gv in the WM of chimpanzees was significantly higher than in macaques, baboons, and humans (all p values $< .05$; Figure 11a). The ANOVA examining GFAPv showed significant main effects of area ($F_{1,21} = 24.68$, $p < .01$) and species ($F_{3,21} = 7.06$; $p < .01$) and a significant interaction ($F_{3,21} = 5.65$, $p < .01$; Figure 11b). LSD post hoc testing showed that chimpanzee GFAPv in the WM was significantly lower than that in macaques or baboons (both p values $< .05$; Figure 11b). Post hoc testing also revealed that macaque WM GFAPv was significantly higher than that of humans ($p < .05$; Figure 11b). Finally, layer I GFAPv of humans was higher than that of all other primates in the study (all p values $< .05$; Figure 11b).

The ANOVA examining the ratio of GFAP/Gv showed significant main effects for both area ($F_{1,21} = 88.45$, $p < .01$) and species ($F_{3,21} = 9.37$, $p < .01$), with a significant interaction ($F_{3,21} = 12.35$, $p = < .01$; Figure 11c). Post hoc analysis showed that chimpanzee WM GFAP/Gv was significantly lower than that of the other species (all p values $< .05$; Figure 11c). GFAP/Gv in layer I of humans was higher than that in macaques, baboons, and chimpanzees (all p values $< .05$; Figure 11c).

The mixed model ANOVA examining GFAP-ir SV showed significant main effects of area ($F_{1,21} = 90.34$, $p < .01$) and species ($F_{3,21} = 19.00$, $p < .01$), with a significant interaction ($F_{3,21} = 6.26$, $p < .01$; Figure 11d). LSD post hoc testing revealed that the GFAP-ir SV of chimpanzee astrocytes in the WM was smaller than the GFAP-ir SV of astrocytes in the WM of macaques, baboons, and humans (all p values $< .05$; Figure 11d). Macaque WM astrocytes had a smaller GFAP-ir SV that was significantly different from human WM astrocytes ($p < .05$; Figure 11d). Human layer I astrocytes had a GFAP-ir SV that was greater than those of macaques, baboons, and chimpanzees (all p values $< .05$; Figure 11d).

3.3 | Astrocyte profiles among species: GLUT1 and EAAT2

There were no significant differences between sexes for any of the astrocyte markers examined; therefore, sexes were pooled for all analyses (Student's t-test, all p values $> .008$). Figure 12 shows EAAT2 immunostaining. There was a significant species-level difference for EAAT2v ($X_3 = 11.75$, $p < .05$; Figure 13a). Post hoc testing demonstrated that humans have greater EAAT2v than baboons and chimpanzees (both p values $< .05$; Figures 12 and 13a). Macaques had greater EAAT2v than chimpanzees ($p < .05$). The one-way ANOVA comparing EAAT2/Gv found a significant difference among species ($F_{3,18} = 4.31$, $p < .05$; Figure 13b). Post hoc testing showed that chimpanzees have increased EAAT2/Gv compared to macaques and baboons (all p values $< .05$; Figure 13b). Additionally, the ANOVA examining EAAT2sv was significant among species ($F_{3,18} = 10.19$, $p < .05$; Figure 13c). Post hoc testing found that chimpanzees have larger soma volumes than all other species, while humans had smaller soma volumes than baboons (all p values $< .05$; Figure 13c).

GLUT1-ir vessel volume was also significantly different among species ($F_{3,21} = 11.14$, $p < .05$; Figure 14a). GLUT1-ir vessels were present in all species, often throughout the upper cortical layers

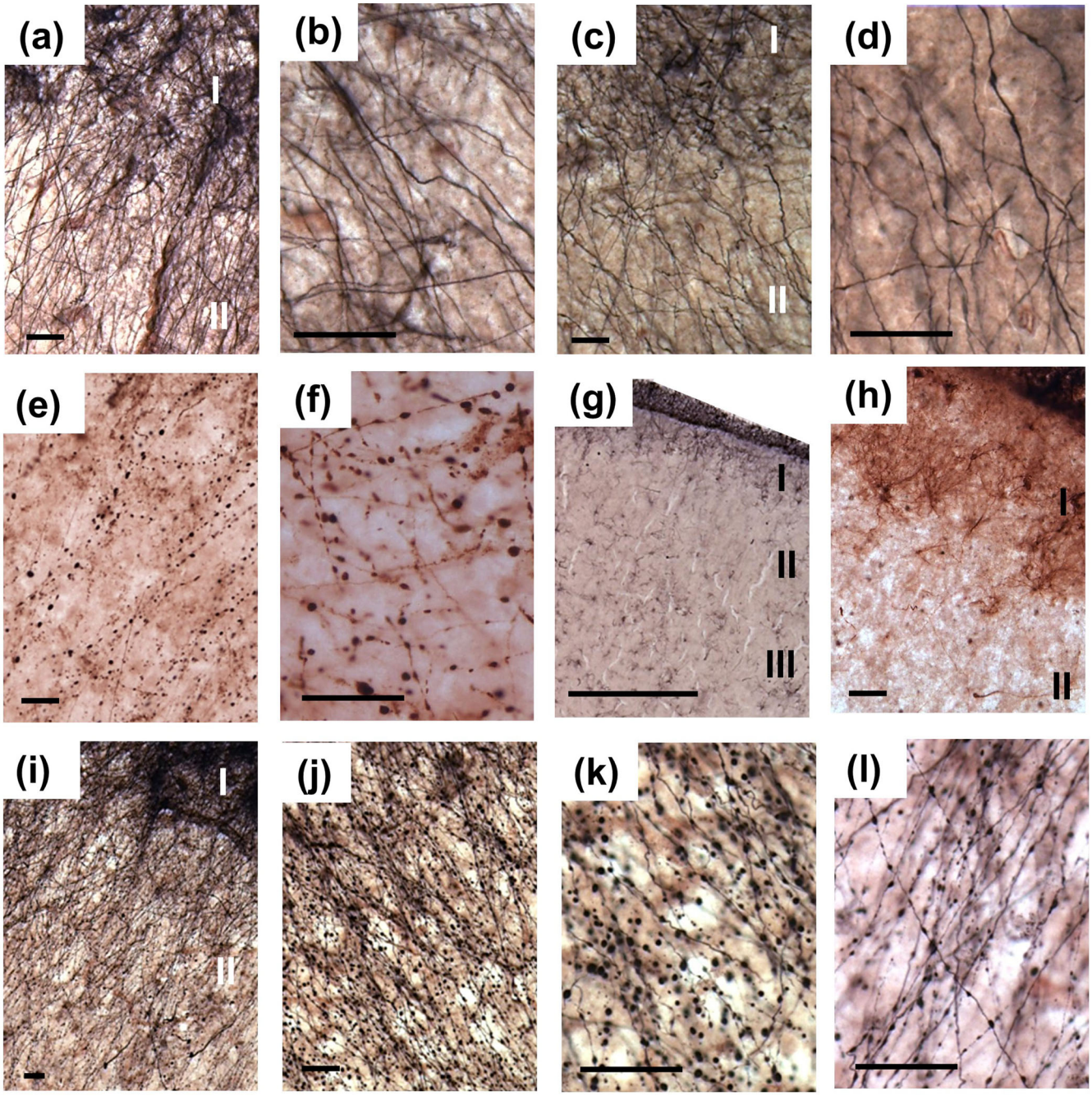


FIGURE 7 Photomicrographs showing glial fibrillary acidic protein (GFAP) immunostaining in layer I of prefrontal cortex area 9 in macaque (a and b), olive baboon (c and d), chimpanzee (e–h), and human (i–l). Both macaques and baboons (a–d) possessed fewer interlaminar processes than humans (i). Chimpanzee interlaminar processes exhibited a bead-like appearance (e and f), indicative of varicosities. Varicosities can enhance the ability of interlaminar astrocytes to modulate neuronal activity. However, the extension of long processes in the form of a palisade was not present in every chimpanzee. Panel g shows the absence of a palisade in a young chimpanzee with no Alzheimer's disease (AD)-type pathology, and panel h shows the loss or absence of a palisade in a chimpanzee with AD-type pathology. Human interlaminar processes were very densely packed (I and j) and, like chimpanzees, had many varicosities on their processes. Scale bars = 25 μm (a–f, h–l) and 250 μm (g).

(Figures 15 and 16). Post hoc analyses showed that humans and macaques do not differ in the volume of GLUT1-ir vessels ($p > 0.05$; Figure 14a). However, both macaques and humans had greater glucose transporter expression than baboons and chimpanzees (all p values $< .05$; Figure 14a). Finally, the ANOVA comparing percent AF of GLUT1-ir vessels showed significant species-level differences ($F_{3,21} = 9.40$, $p < .05$; Figure 14b). Post hoc tests revealed that both macaques and humans have AF of GLUT1-ir vessels greater than either baboons or chimpanzees (all p values $< .05$; Figure 14b).

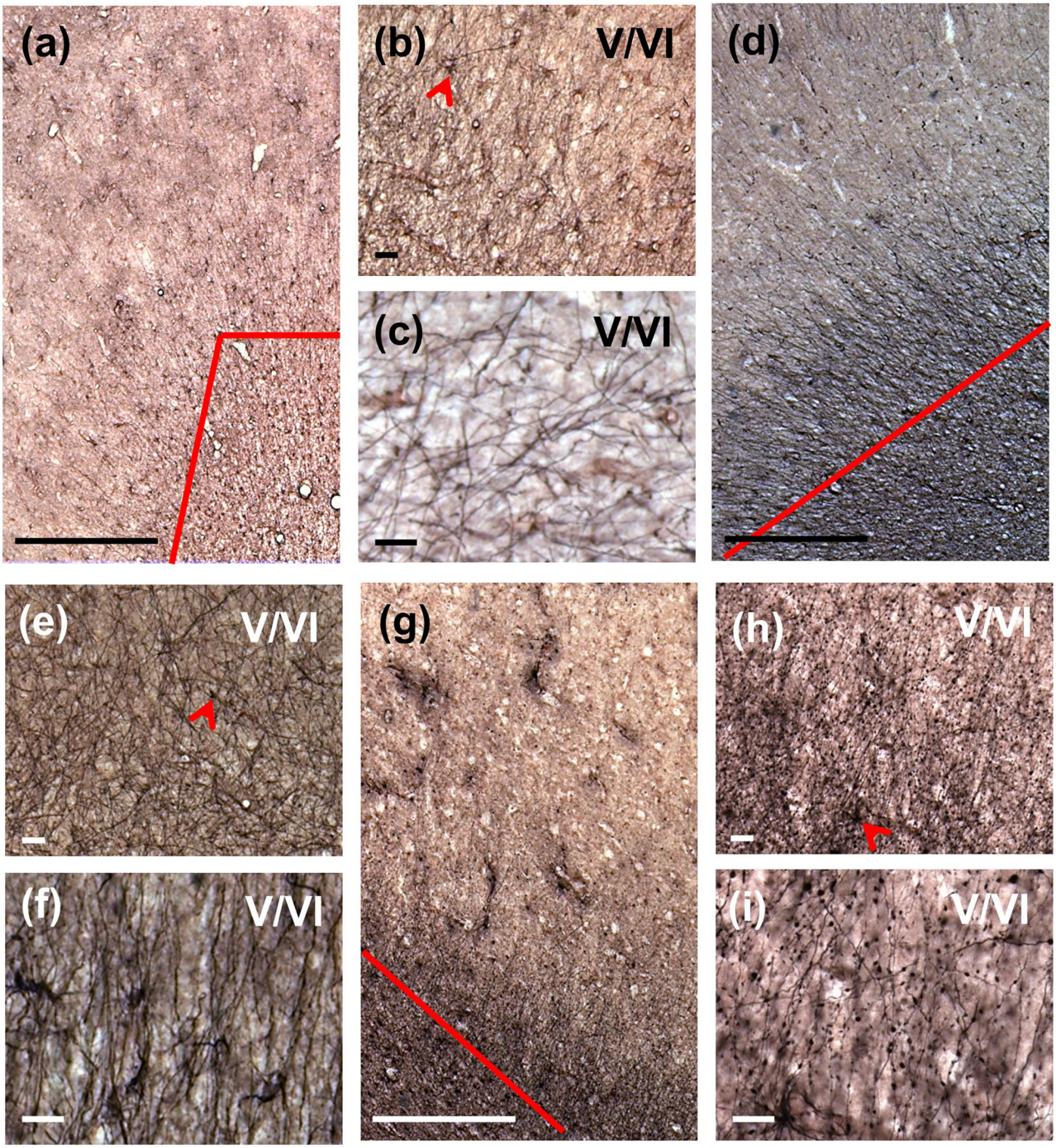


FIGURE 8 Glial fibrillary acidic protein (GFAP) immunostaining in the deeper cortical layers of prefrontal cortex area 9 of macaque (a–c), olive baboon (d–f), and human (g–i). The red line demarcates the border between white matter and gray matter (a, d, g). The red arrows indicate examples of GFAP-ir astrocyte cell bodies (b, e, and h). Astrocyte processes are found in the deeper cortical layers near the white matter (layers V and VI) and extend toward the upper cortical layers (panels a, d, and g). These processes parallel the interlaminar processes found in the upper cortical layers. Most of these processes extend from cell bodies in layer V/VI (b, e, and h); however, some processes extend from the white matter. Qualitatively, the number of upwardly extending processes increases from macaques (a) with the fewest and then baboons (d) and humans (g) with the densest palisade. Lastly, these processes have a similar torturous appearance often described for interlaminar processes (c and f). Varicosities were occasionally found in nonhumans; however, humans routinely possessed varicosity-like processes. Scale bars = 25 μm (b, c, e, f, h, and i) and 250 μm (a, d, and g).

TABLE 4 Descriptive statistics for all variables with the mean and standard deviation for each species

Variable	*Macaca nemestrina*	*Papio anubis*	*Pan troglodytes*	*Homo sapiens*
Total glia density layer I (Gv, mm^3)	102,717 ± 25,444 (6)	100,834 ± 13,036 (6)	120,856 ± 63,414 (7)	75,093 ± 13,712 (6)
Total glia density WM (Gv, mm^3)	138,788 ± 26,692 (6)	124,734 ± 14,266 (6)	267,012 ± 163,862 (7)	120,832 ± 21,479 (6)
GFAP-ir astrocyte density layer I (GFAPv, mm^3)	8,901 ± 1,120 (6)	8,071 ± 1,521 (6)	7,662 ± 3,735 (7)	12,496 ± 3,088 (6)
GFAP-ir astrocyte density WM (GFAPv, mm^3)	9,016 ± 2,124 (6)	6,503 ± 1,078 (6)	2,778 ± 1,948 (7)	5,333 ± 2,905 (6)
GFAPv/Gv layer I	0.09 ± 0.02 (6)	0.08 ± 0.02 (6)	0.08 ± 0.05 (7)	0.17 ± 0.04 (6)
GFAPv/Gv WM	0.07 ± 0.02 (6)	0.05 ± 0.01 (6)	0.01 ± 0.003 (7)	0.05 ± 0.02 (6)
GFAP-ir astrocyte soma volume layer I (GFAPSV, μm^3)	356 ± 25 (6)	431 ± 56 (6)	388 ± 116 (7)	597 ± 89 (6)
GFAP-ir astrocyte soma volume WM (GFAPSV, μm^3)	274 ± 45 (6)	349 ± 85 (6)	157 ± 18 (7)	418 ± 63 (6)
EAAT2-ir astrocyte density WM (EAAT2v, mm^3)	50,171 ± 5,289 (6)	41,323 ± 3,390 (6)	37,966 ± 13,139 (4)	55,368 ± 7,799 (6)
EAAT2-ir astrocyte soma volume WM (EAAT2SV, μm^3)	100 ± 12 (6)	118 ± 15 (6)	146 ± 21 (4)	91 ± 18 (6)
GLUT1-ir vessel volume (mm^3)	205,570,265 ± 52,011,586 (6)	106,007,577 ± 37,699,932 (6)	141,516,607 ± 38,657,198 (7)	222,333,218 ± 29,751,314 (6)
GLUT1-ir vessel area fraction (%)	11 ± 2 (6)	6 ± 2 (6)	8 ± 3 (7)	12 ± 1 (6)

Note: The number of individuals analyzed is provided in parentheses for each variable.
Abbreviation: WM, white matter.

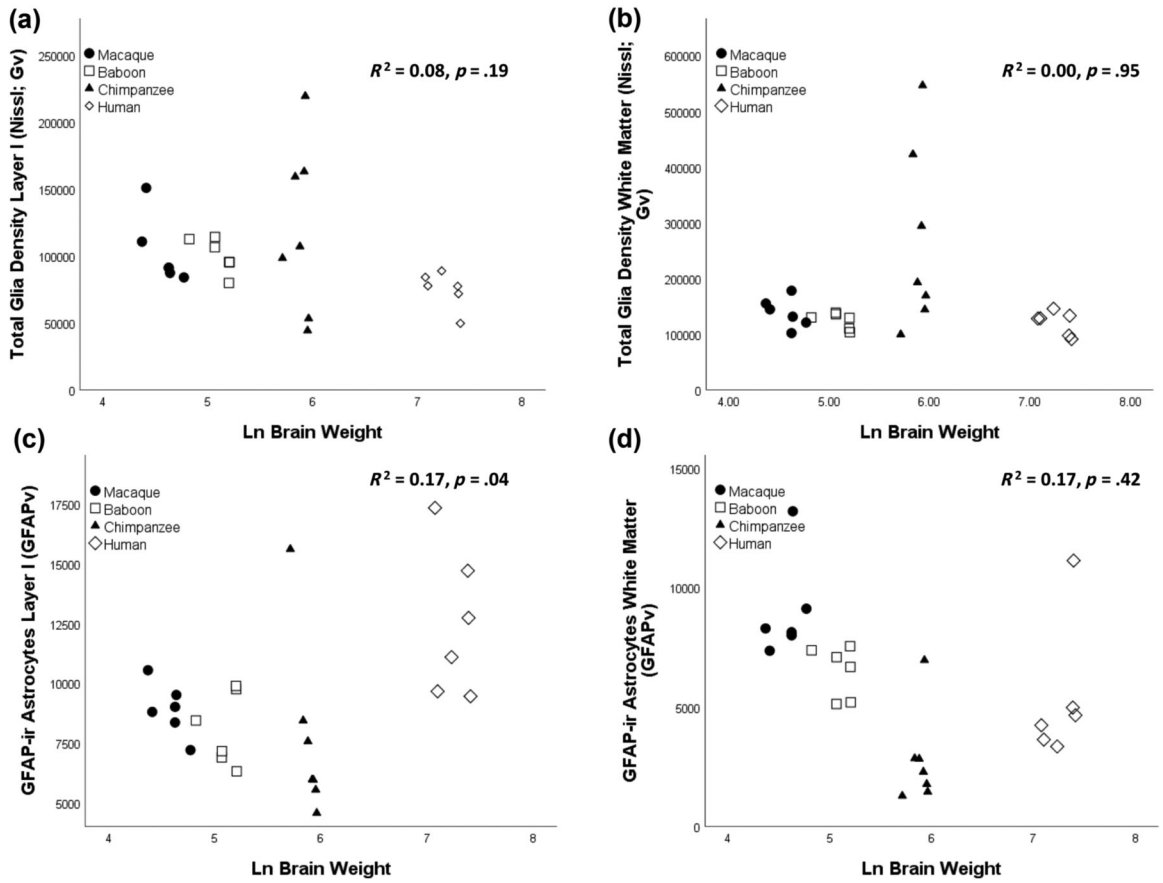


FIGURE 9 Linear regression analyses comparing total glia density (Gv) and glial fibrillary acidic protein (GFAP)-ir astrocyte density (GFAPv) in layer I and white matter by brain weight. The analyses revealed that only layer I GFAPv increased with brain weight. All R^2 values for each individual analysis are provided in the corresponding panel.

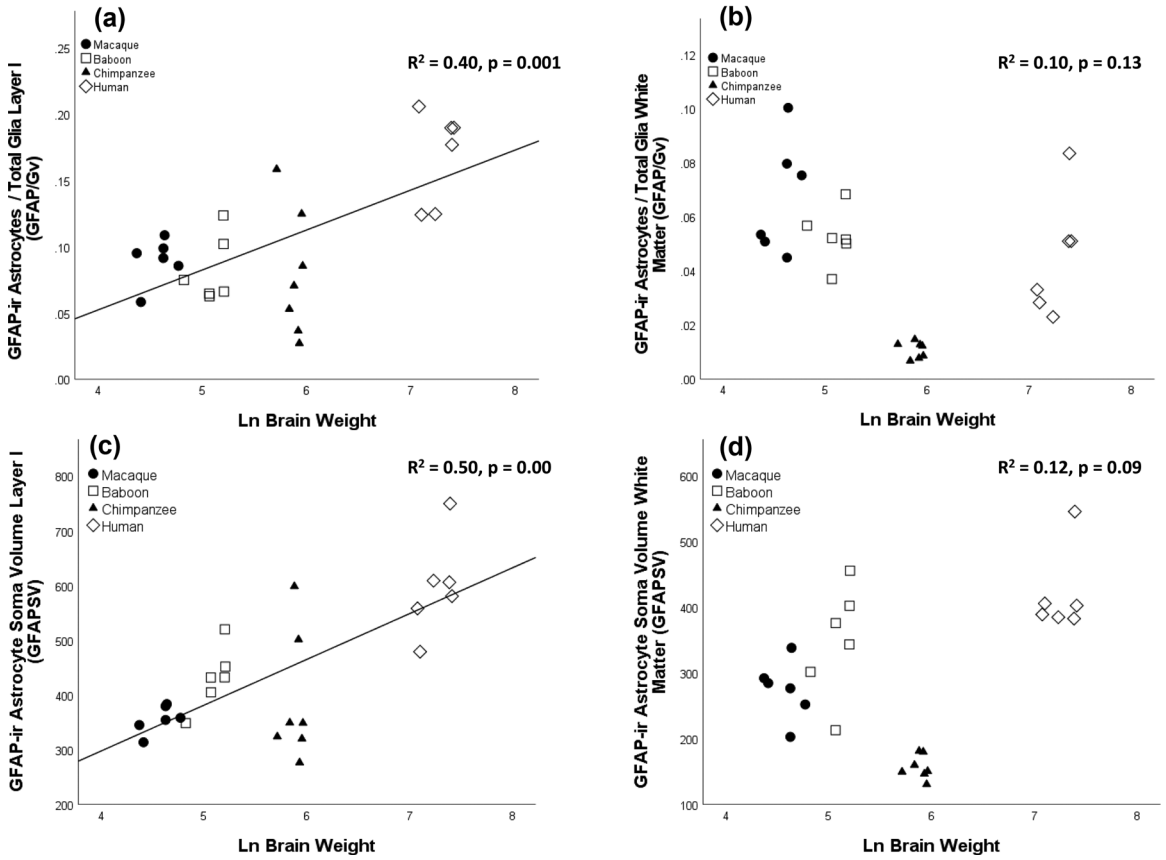


FIGURE 10 Linear regression analyses comparing the ratio of glial fibrillary acidic protein (GFAP)-ir astrocyte density (GFAPv) to total glia density (Gv) in layer I and white matter by ln brain weight (a and b). The analyses revealed that this ratio was significant in layer I ($p < .05$). Panels c and d show the linear regression of GFAP-ir astrocyte SV with ln brain weight in layer I and white matter. Soma volume increased with brain weight in layer I ($p < .05$). All R^2 values for each individual analysis are provided in the corresponding panel.

4 | DISCUSSION

We found an increase in the ratio of GFAP-ir astrocytes/total glia and GFAP-ir astrocyte soma volume in layer I with brain weight. Interlaminar astrocytes occupy layer I and typically extend one or two processes into deeper cortical layers (Colombo, 1996; Colombo et al., 1995; Colombo & Bentham, 2006; Colombo & Reisin, 2004; Falcone et al., 2019; Oberheim et al., 2006, 2009). Therefore, even though their cell bodies are in layer I, these astrocytes influence the homeostatic environment of more neuronally rich and metabolically demanding regions. Hence, if larger brains have increased numbers of interlaminar astrocytes, it could contribute to more efficient metabolic processing. Humans and chimpanzees were the only two species that possessed varicose interlaminar processes. This is consistent with previous comparative studies of primates demonstrating that varicose projection astrocytes are found only in hominoids (Falcone et al., 2022). A single human astrocyte has been estimated to make contact with 270,000 to 2 million synapses, where it has the potential to release several neuroactive molecules, such as glutamate, D-serine, ATP, adenosine, γ-aminobutyric acid, tumor necrosis factor-α, and prostaglandins—all factors that can influence neuronal and synaptic physiology (Bushong et al., 2002; Oberheim et al., 2009; Volterra & Bezzi, 2002). A study of neuronal axon varicosities demonstrated that varicosity density is related to the activity of surrounding pyramidal cells in the prefrontal cortex (Z. W. Zhang et al., 2011). Therefore, we expect that the presence of varicosities on the projecting interlaminar palisades in humans, chimpanzees, and other hominoids correlates with increased interconnectivity of these astrocytes.

Chimpanzees differed from all other species, with a greater WM glia population. The average chimpanzee area 9 WM glial density was 287,469 per mm^3 compared to humans with 120,833 per mm^3. A recent study of glial subtypes in chimpanzees found that oligodendrocytes made up 81% of the total glia, with 8% being astrocytes and 11% microglia (Edler et al., 2020). Therefore, the increase in chimpanzee WM glia density is likely due to an increase in oligodendrocytes. This would also account for GFAPv/Gv being lower in chimpanzees, as it would reflect the higher total glial density in the species.

Humans showed higher densities of GFAP-ir interlaminar astrocytes (layer I) with significantly larger soma volumes. Notably, the percentage of GFAP-ir astrocytes in layer I of humans was 17% compared with 8–9% observed for nonhuman primates. Humans have the largest brains of any primates, and while the human brain comprises 2% of the body's total mass, it utilizes approximately 20% of

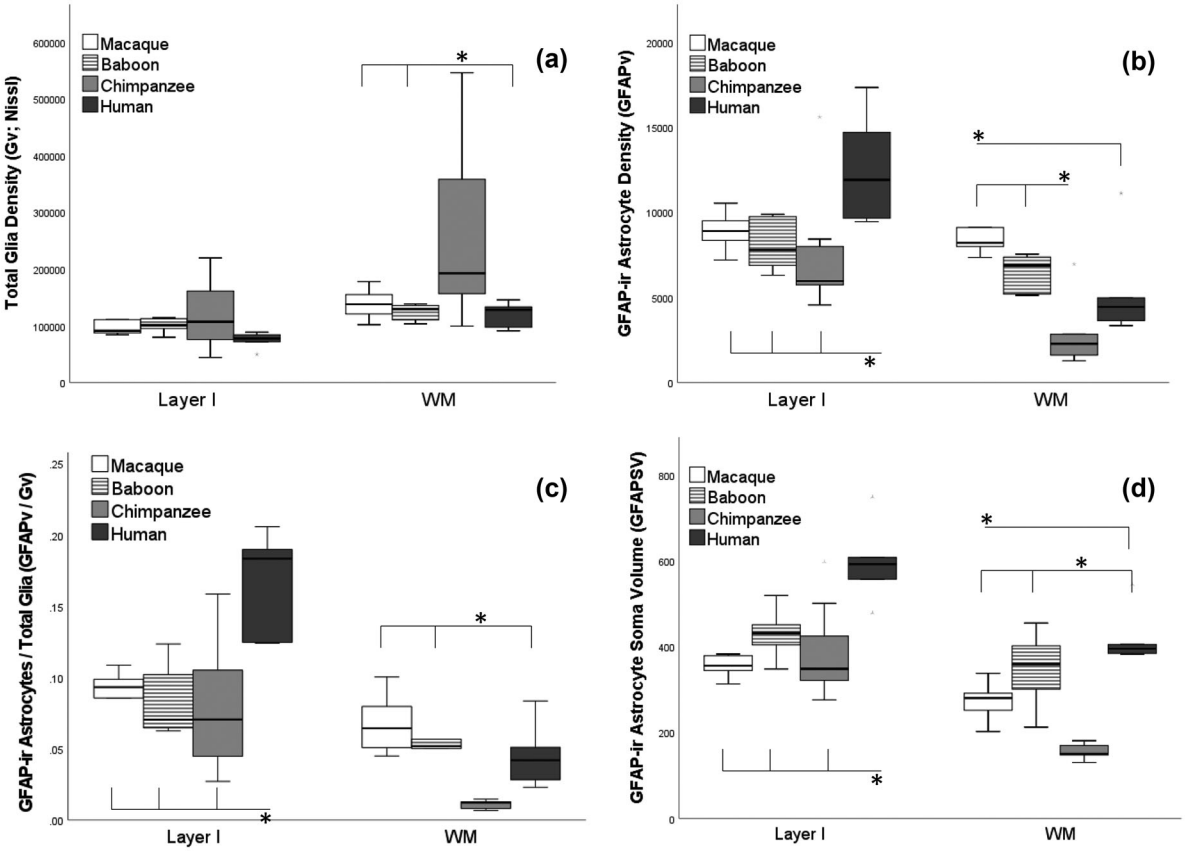


FIGURE 11 A repeated-measures analysis of variance (ANOVA) was performed to analyze glia density (Gv), glial fibrillary acidic protein (GFAP)-ir astrocyte density (GFAPv), GFAP/Gv, and GFAP-ir soma volume (GFAPSV) by region and species. Gv in chimpanzees was found to be significantly greater in the white matter compared to the other three primate species ($p < .05$, a). Humans had a significantly greater layer I GFAPv (b), GFAP/Gv (c), and GFAPSV (d) (all p values $< .05$). In the white matter, macaques had a significantly greater GFAPv than chimpanzees and humans, while baboons were significantly greater than chimpanzees (p values $< .05$; b). Chimpanzees had a significantly lower ratio of GFAP/G (c) and decreased GFAPSV (d) in the white matter compared to the other three primate species (p values $< .05$). Humans also had a greater GFAPSV in the white matter than macaques and chimpanzees (p values $< .05$; d). Asterisks denote statistically significant differences.

the body's total metabolic costs (Sokoloff, 1960). In comparison, non-human primate brains utilize approximately 9–12% of the body's total energy requirements (Mink et al., 1981; Sokoloff, 1960). More recent studies have demonstrated that the human neocortex differs in having an increased expression of metabolic energy genes (Bauernfeind et al., 2015; Cáceres et al., 2003; Fu et al., 2011; Grossman et al., 2004; Khaitovich et al., 2006; Preuss et al., 2004; Uddin et al., 2004; Vasile et al., 2017). Furthermore, analyses have revealed that only 30% of the genes enriched in human astrocytes are similarly enriched in rodents (Vasile et al., 2017). These upregulated astrocytic genes are related to their functional roles in metabolism, calcium signaling, synaptogenesis, and neurite growth (Vasile et al., 2017). Since astrocytes play a predominate role in metabolism, their subsequent increase in number and size in layer I of the human prefrontal cortex would allow the expanded human neocortex to efficiently utilize energy sources.

Along with its role in metabolic processing, the astrocyte–neuron lactate shuttle has been shown to play a role in memory formation (Suzuki et al., 2011). Several lines of research have also implicated the important role of astrocytes in learning and memory (Han et al., 2013; Suzuki et al., 2011). Therefore, an increase in GFAP-ir astrocyte density in humans may play a role in our unique cognitive ability. Interlaminar astrocytes are also uniquely susceptible to neurodegenerative disease (Colombo et al., 2002; Colombo & Reisin, 2004; Munger et al., 2019). Therefore, if humans have a greater density of interlaminar astrocytes and more elaborate and developed palisades, which are intimately integrated into the homeostatic regulation of deeper cortical layers, their loss due to disease could be extremely detrimental. Loss of interlaminar astrocytes in humans could result in the breakdown of long-distance astrocyte–neuron communication, resulting in disruptions in metabolic processing, ion reuptake, and glial and neurotransmitter signaling at a scale not seen in other species.

The major energetic burden in the brain occurs at the synapse, and the astrocyte–neuron lactate shuttle couples synaptic activity (energy use) to an energy delivery system (Halestrap, 2013; Magistretti et al., 1999; Magistretti, 2006; Magistretti & Pellerin, 1999; Pellerin et al., 1998; Pellerin & Magistretti, 1994, 1997; Pierre & Pellerin, 2005). Two major components of this system are EAAT2 and GLUT1 (Pellerin et al., 2007; Pellerin & Magistretti, 1994, 1997). An increased density of EAAT2 is hypothesized to be an indicator of the greater metabolic

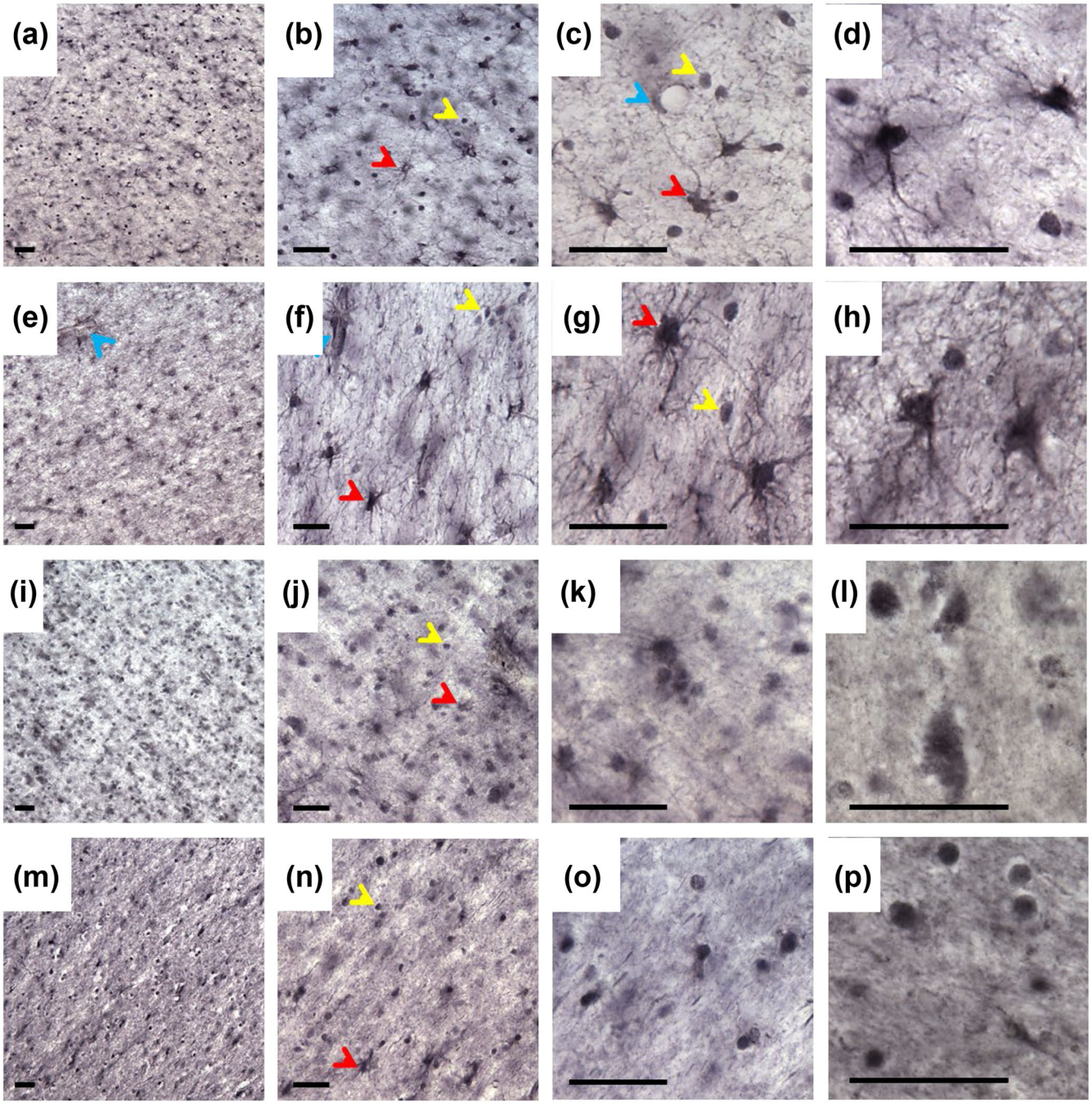


FIGURE 12 Photomicrographs of sections immunostained for excitatory amino acid transporter 2 (EAAT2) in macaques (a–d), baboons (e–f), chimpanzees (e–l), and humans (m–p). EAAT2 cells and processes are visualized at lower magnification (a, b, e, g, i, j, m, and n) and higher magnification (c, d, g, h, k, l, o, and p). EAAT2-ir cells had two distinct morphological subtypes. The first subtypes (i.e., yellow arrows) consisted of a round cell soma and limited number of processes. This cell type appeared to predominate in every species. The second subtype (i.e., red arrows) had a pyramidal cell soma and an increased number of processes. These cells appeared less frequently overall but did appear to be more common in macaques and baboons. The processes of EAAT2-ir cells often interacted with blood vessels (blue arrows; c, e, and f). Scale bars = 25 μm.

demand, as transport of glutamate into astrocytes is an initial trigger for glucose uptake and utilization (Pellerin & Magistretti, 1994). EAAT2 also plays a primary role in glutamate signal transduction, as it is responsible for the rapid termination of glutamate signaling and the maintenance of extracellular glutamate levels (Shigeri et al., 2004). GLUT1, located on astrocytes and capillary endothelial cells, is responsible for the uptake of circulating glucose (Magistretti & Allaman, 2015; Pellerin et al., 2007; Simpson et al., 2007; Winkler et al., 2015). Our study found that humans had greater EAAT2v and EAAT2/Gv than chimpanzees and baboons, but humans did not significantly differ from

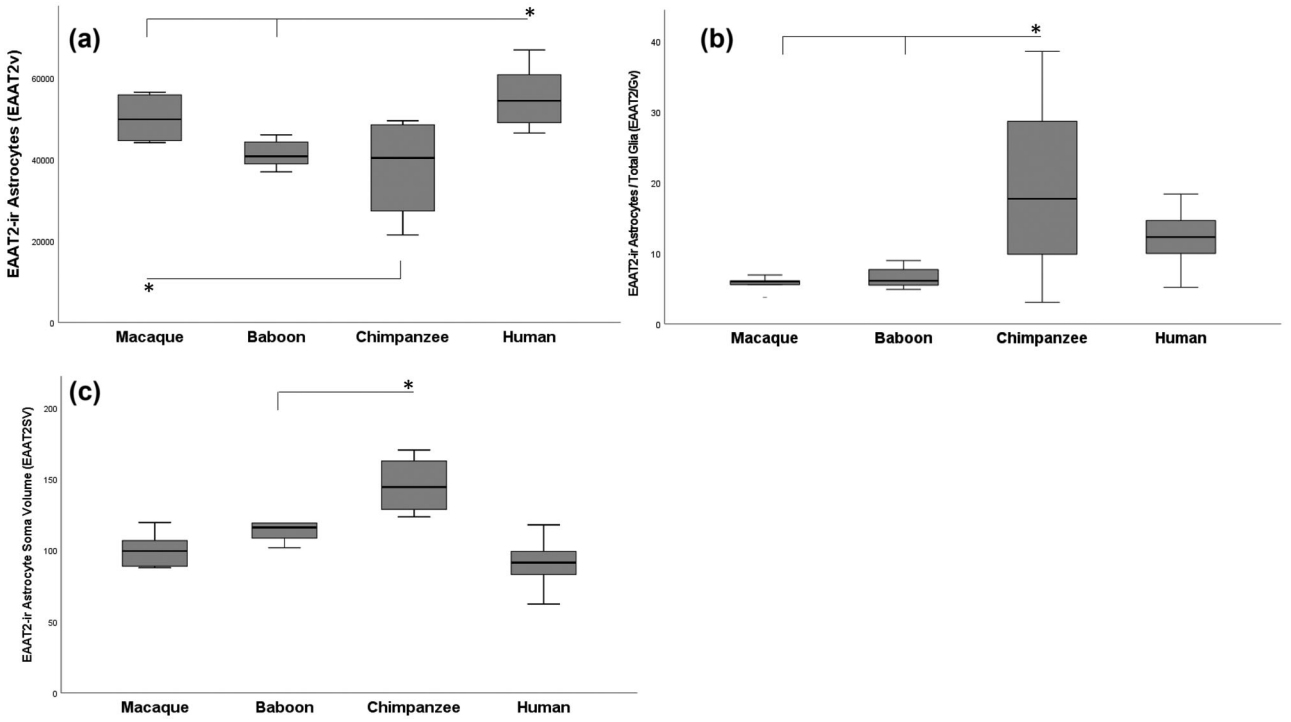


FIGURE 13 Kruskal–Wallis analysis was performed for EAAT2v (a), and one-way analysis of variance (ANOVA) was performed for EAAT2v/Gv (b) and EAAT2SV (c). Post hoc testing revealed that humans had significantly greater numbers of EAAT2v (a) and EAAT2v/Gv (b) than chimpanzees and baboons. Macaques also had significantly greater EAAT2v (a) and EAAT2v/Gv (b) than chimpanzees. Post hoc testing also demonstrated that baboons had increased EAAT2v/Gv (b) than chimpanzees. Lastly, chimpanzees had significantly greater EAAT2 SV (c) than all other catarhine primates (macaques, baboons, humans). Asterisks denote statistically significant differences.

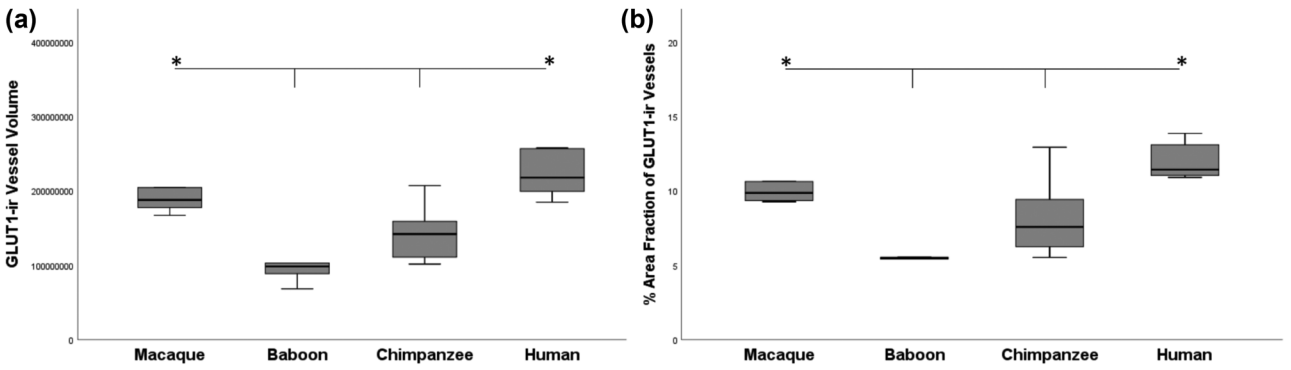


FIGURE 14 A one-way analysis of variance (ANOVA) was performed for glucose transporter (GLUT)-ir vessel volume (a) and percentage of area fraction of GLUT-ir vessels (b). Post hoc testing revealed that humans and macaques shared increased GLUT-ir vessel volume and area fraction relative to baboons and chimpanzees (all p values $< .05$). Asterisks denote statistically significant differences.

macaques. We found the same results for GLUT1-ir vessel volume and area fraction. In this study, the average human brain weighed 1,442 g compared to macaques with an average brain weight of 99 g. Therefore, neither EAAT2 nor GLUT1 appears to be associated with the increased metabolic demands related to the expansion in brain size, which we initially hypothesized.

The synapse is the point in the CNS that causes the greatest energetic burden, and human neocortical neurons consume nearly three times more ATP per cell to maintain resting membrane potential and to fire an action potential compared to rodents (Lennie, 2003). As EAAT2 was quantified in the WM, further research is needed to determine whether EAAT2 differs between humans and macaques in more synaptic-rich layers of the cortex. However, GLUT1-ir was quantified through the gray matter of prefrontal cortex area 9 and should be representative of an energetically demanding region. Overall, GLUT1 immunoreactivity did not differ between humans and macaques, but

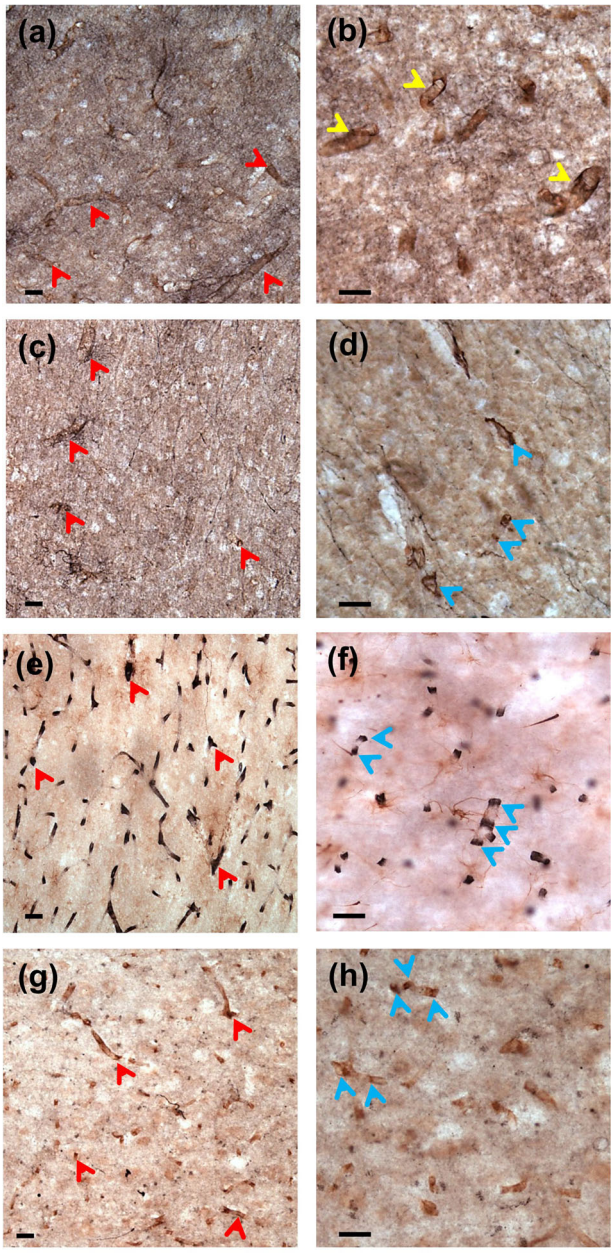


FIGURE 15 Colocalization of glucose transporter 1 (GLUT1) and glial fibrillary acidic protein (GFAP) in the upper cortical layers of macaques (a and b), baboons (c and d), chimpanzees (e and f), and humans (g and h). GLUT1 is labeled in red, and GFAP is labeled in black for macaques (a and b), baboons (c and d), and humans (g and h). For chimpanzees (e and f), GLUT1 is labeled black, and GFAP is labeled in red. GLUT1 stained numerous vessels (red arrows) throughout the upper layers of prefrontal cortex area 9 of each catarrhine primate (a, c, e, and g). GLUT1 often homogenously stained the entire blood vessel in macaques (b; yellow arrows). Alternatively, the blood vessels of baboons (d), chimpanzees (f), and humans (h) had GLUT1 staining that frequently only occurred in small regions (blue arrows), leaving regions of the same blood vessels negative for GLUT1 immunoreactivity. Scale bars = 25 μm.

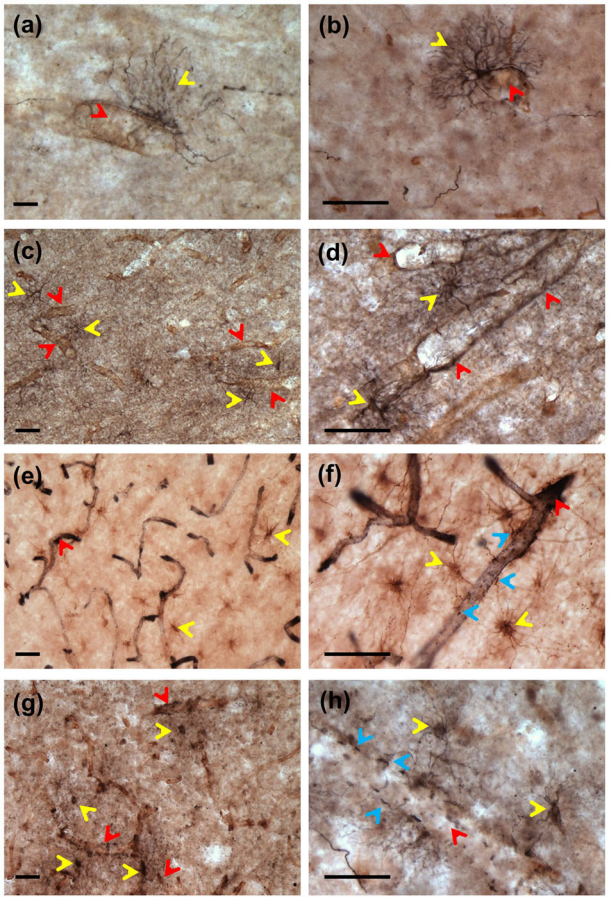


FIGURE 16 Photomicrographs showing colocalization of glucose transporter 1 (GLUT1) and glial fibrillary acidic protein (GFAP) in macaques (a and b), baboons (c and d), chimpanzees (e and f), and humans (g and h) in the upper layers of the prefrontal cortex area 9. GLUT1 is labeled in red, and GFAP is labeled in black for macaques (a and b), baboons (c and d), and humans (g and h). Colors are reversed in chimpanzees (g and h). GFAP-ir astrocytes (yellow arrows) were found to interact with GLUT1-ir blood vessels (red arrows) in every species. Chimpanzees and humans appeared to have more astrocyte-vasculature interactions, and the endfeet of astrocytes (blue arrows) were more easily visible in these species (f and h). Scale bars = 25 μm.

human GFAP-ir astrocytes were often found to be more interactive with the vascular walls of GLUT1-ir vessels. Therefore, humans may be able to more rapidly upregulate glucose intake than macaques, although further colocalization studies are needed.

We did not find a significant correlation between total glial density and brain weight for either layer I or WM. Layer I, the molecular layer, is occupied by the dendrites of cells whose somata are located in deeper cortical layers and axons that are traveling to different cortical regions, while WM contains myelinated axon tracts (Broca, 1878; Brodmann, 1909; Ramón y Cajal, 1897; Pessoa & Hof, 2015). Glial cells are found in abundance in both regions. The presence of neuronal cell bodies is limited in both layer I and the WM, suggesting a local decrease in neuronal energy demands. Importantly, while glia to neuron ratios have been associated with increased brain size, that association is

the result of variable neuron densities, whereas glia densities are more invariant across brain sizes of mammals (Dos Santos et al., 2020; Herculano-Houzel, 2014).

In conclusion, and in concordance with previous reports, the overall glia density per unit volume is consistent across species and is not associated with brain size in our sample of primates, including humans (e.g., Herculano-Houzel, 2014). Similarly, EAAT2-ir and GLUT1-ir are not related to brain expansion. Remarkably, layer I astrocyte densities and soma size are increased in humans, nearly double those of the nonhuman primates in our sample. It is interesting to note that layer I astrocytes possess a unique vulnerability to neurodegenerative processes, including AD (e.g., Colombo et al., 2002; Munger et al., 2019). However, these increases may translate into more efficient metabolic processing and increased cognitive functioning for the human brain.

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AUTHOR CONTRIBUTIONS

Emily L. Munger and Mary Ann Raghanti conceptualized the study. Emily L. Munger, Melissa K. Edler, Patrick R. Hof, Chet C. Sherwood, William D. Hopkins, and Mary Ann Raghanti obtained the brains. Emily L. Munger, Melissa K. Edler, and Mary Ann Raghanti processed the specimens. Emily L. Munger performed staining and analysis. Emily L. Munger and Mary Ann Raghanti wrote the manuscript, and all authors contributed to editing the manuscript.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

All data have been presented in this manuscript and are available in Excel or SPSS files upon request to Emily L. Munger or Mary Ann Raghanti.

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